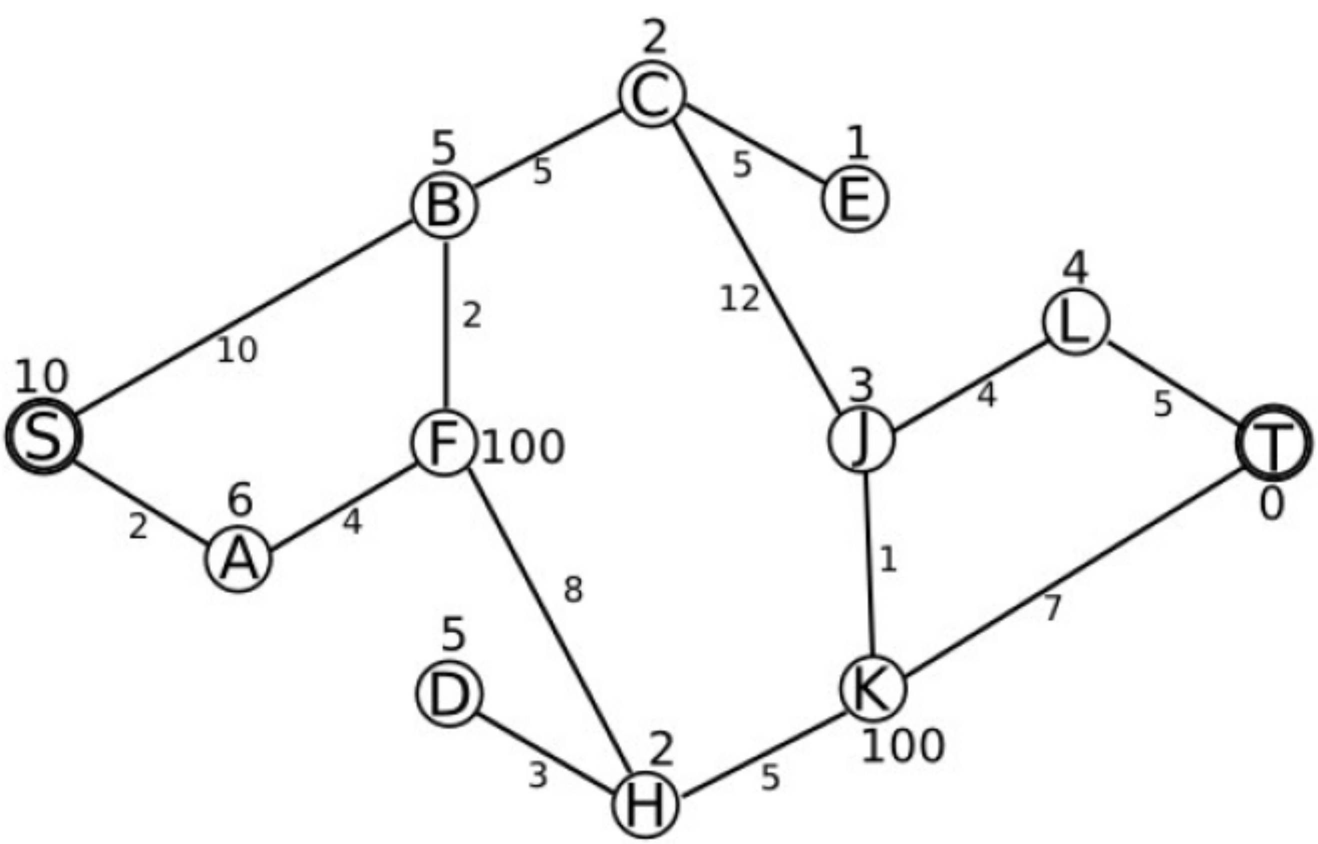
Practice Problems:

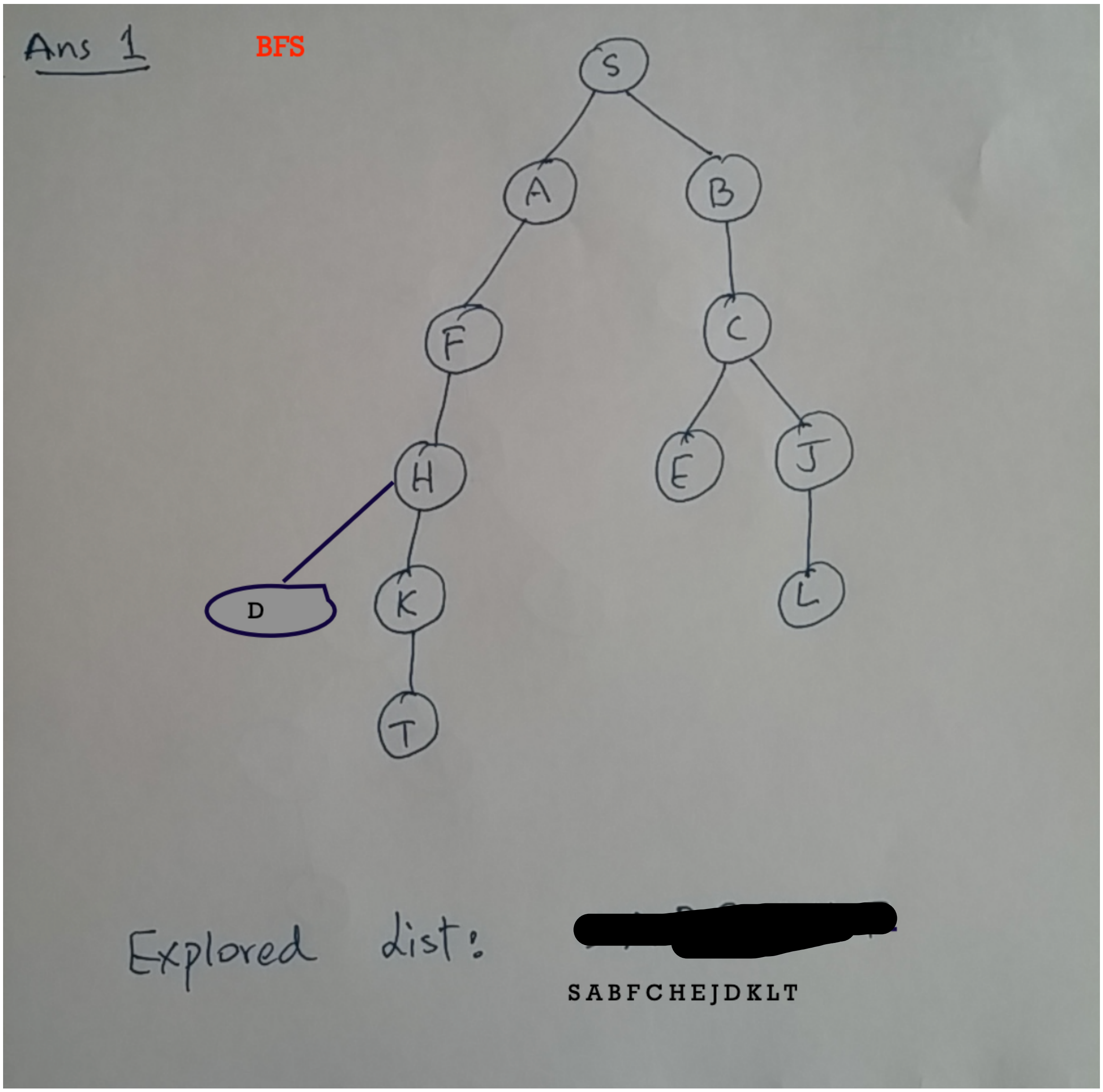
Question 1: You have moved to a new city and are trying to learn how to get from your home at **S** to the train station at **T.** Some streets have more traffic than others. Your friend who has lived in this city for a long time provides you with information about the traffic on each street – the streets are labeled with costs, in the form of how many minutes it will take you to traverse each one. You also have access to a distance heuristic, which tells you the estimated distance from each node to T. The distance is written above or below each node. (Note: You will not use the heuristic in uninformed search strategy)

>>>In all search problems, use alphabetical order to add elements to the frontier
>>>Use Graph Search for all questions.

Q1: Use Breadth First Search to find the path from S to T. (1) Draw the part of the search tree that is explored by the search. (2) List in order all nodes/states inserted into the explored list.
Q2: Use Uniform cost to find the path from S to T. (1) Draw the part of the search tree that is explored by the search. (2) List in order all nodes/states inserted into the explored list.

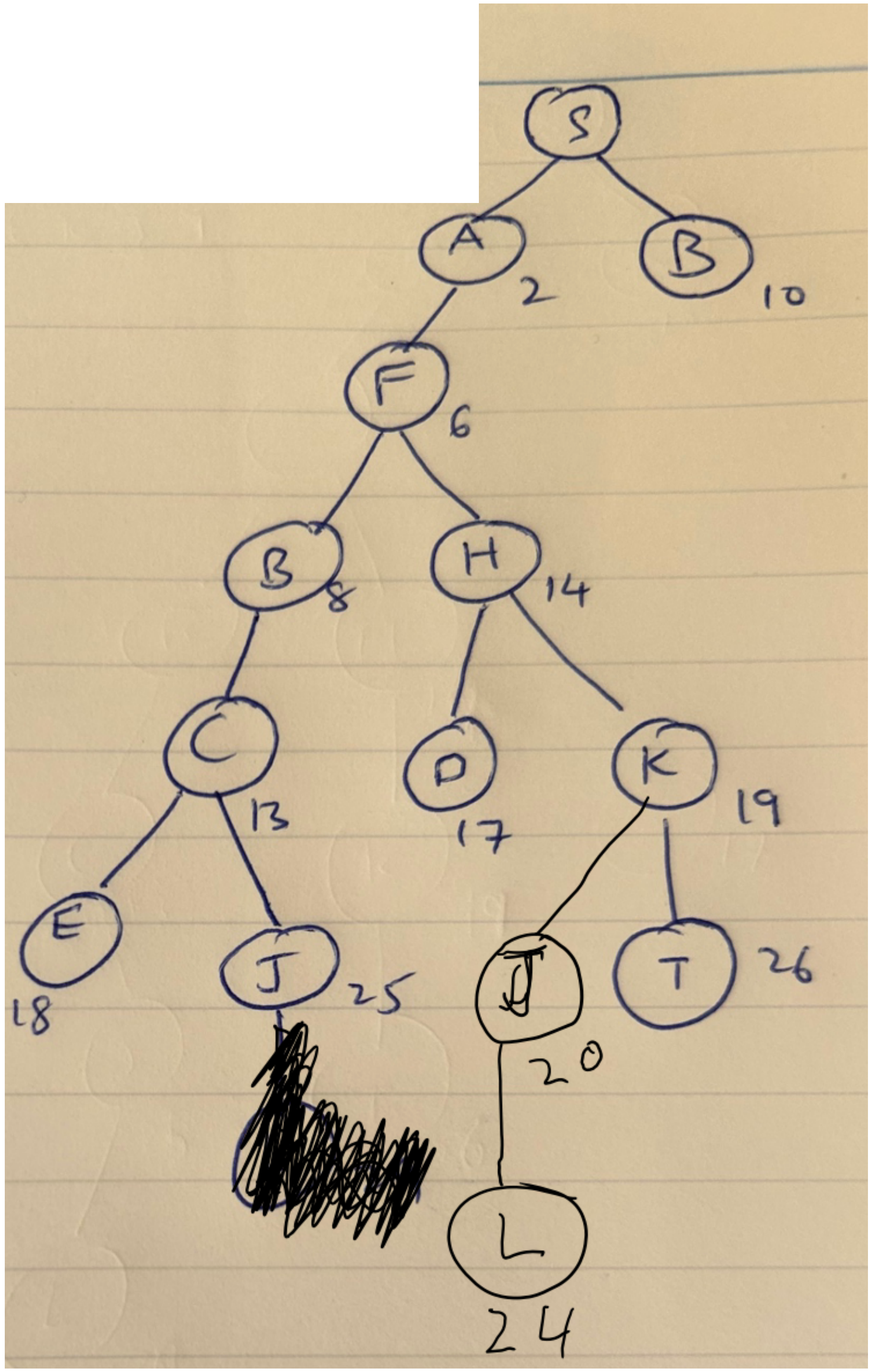


Solution:


Ans 1
BFS
S
A
B
F
C
H
E
J
D
K
L
T
Explored list: [redacted]
SABFCHEJDKLT

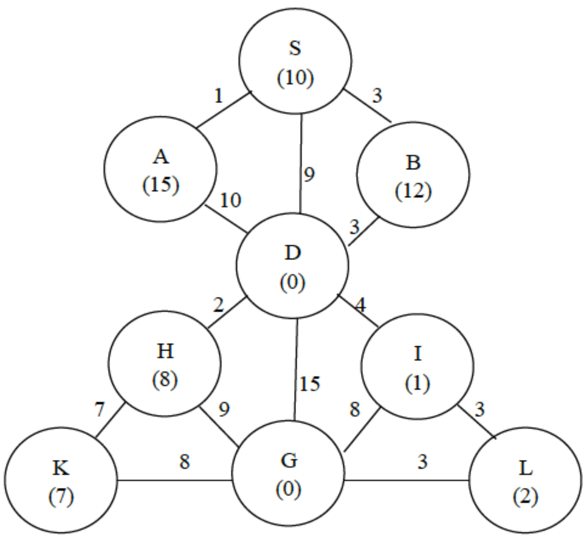
Uniform cost search

Explored: SAFBCHGEKJLT

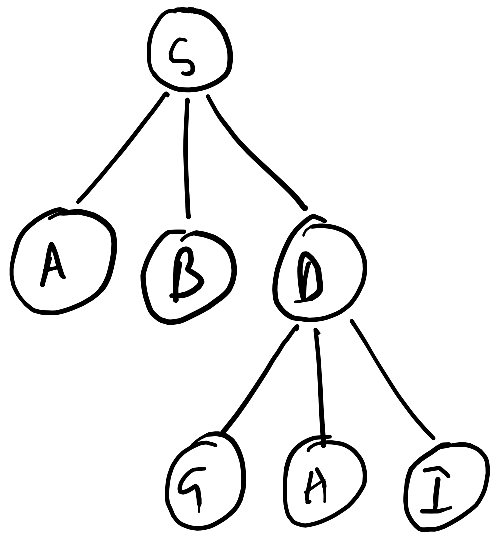


Question 2: Start state S, Goal state G.
>>>In all search problems, use alphabetical order to break ties when deciding the priority to use for removing paths from frontier.
>>>Use Graph Search for all questions.

Q1: Use Breadth First Search to find the path from S to G. (1) Draw the part of the search tree that is explored by the search. (2) List in order all nodes/states inserted into the explored list.
Q2: Use Uniform cost to find the path from S to G (1) Draw the part of the search tree that is explored by the search. (2) List in order all nodes/states inserted into the explored list.



BFS



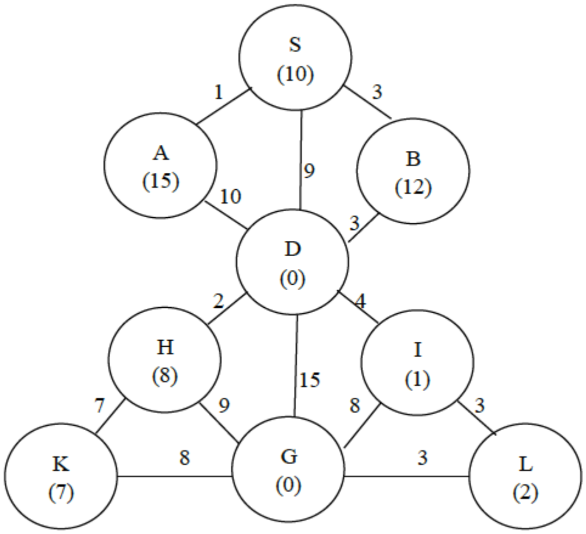
Frontier

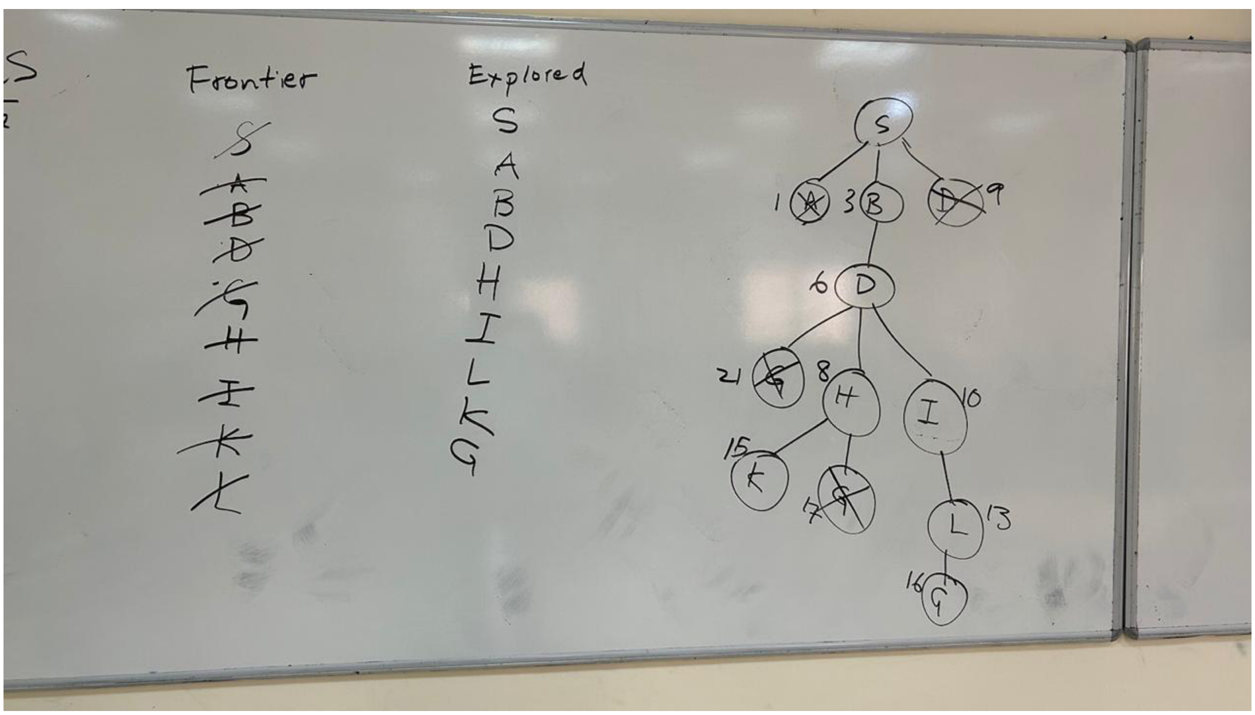
~~S~~
~~A~~
~~B~~
~~D~~
~~G~~
H
I

Explored

SABDG

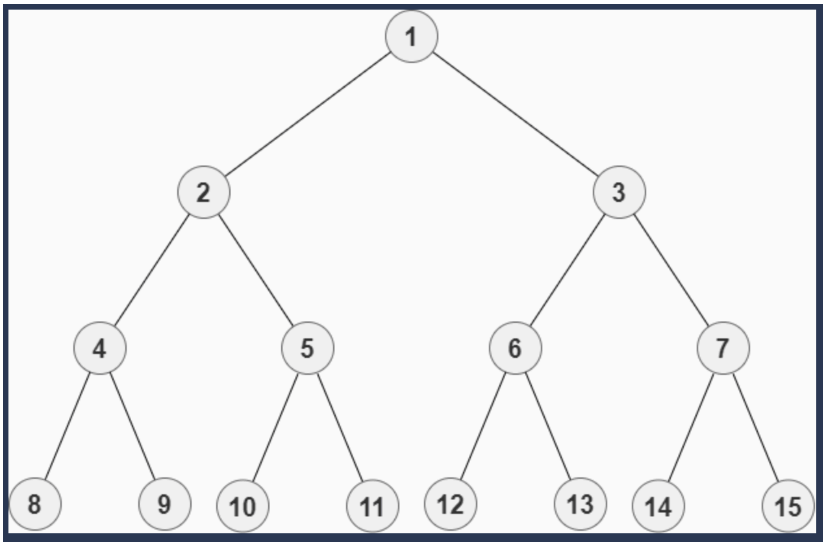
Uniform cost





Question 3: Consider a state space where the start state is number 1 and each state k has two successors: numbers 2k and 2k + 1.

1. Draw the portion of the state space for states 1 to 15.



2. Suppose the goal state is 11. List the order in which nodes will be visited for breadth-first search, depth-limited search with limit 3, and iterative deepening search.

Breadth-first search would visit nodes in this order: 1 2 3 4 5 6 7 8 9 10 11

Depth-limited search (with limit 3) would visit nodes in this order: 1 2 4 8 9 5 10 11

Iterative deepening search would visit nodes in this order: 1 1 2 3 1 2 4 5 3 6 7 1 2 4 8 9 5 10 11